CSE 331/EEE 453: Microprocessor Interfacing and Embedded Systems

Assignment#3: Microprocessor Interfacing

Summer 2025

Name: ____________________________________

Student ID: ___________________________

Direction: **You can consult any resource such as books, online references, and videos for this assignment; however, you have to properly cite and paraphrase your answers when it is necessary. There will be points for partial attempts. There will be three days of grace period for late submission; however, 20% of points will be deducted.**

Submission guidelines: **All our assignments will be based on 100 points so that we can assign weights to get your points for final grade. Whoever fails to submit within the time period assigned through NSU canvas but submit the assignment within the next three days will be penalized by 20% deduction of points (e.g. start with -20 points within 100 points). The late submission will be open until semester to your TA. After the first 3 days of grace period, the deduction will be 30%.**

Note: **You are allowed to use only instructions implemented by the actual ARM hardware. Note, base ten numbers are listed as normal (e.g. 23), binary numbers are prefixed with 0b or format such as XX_2 and hexadecimal numbers are prefixed with 0x or format such as XX_{16}/XX_{HEX}**

1. **Problem (100 points):** You have to write programs to find the prime numbers from the following integers: 4, 7, 11, 14, 16, 21, 23, 27, 29, 38, 39, 41, 47, 49, 53, 55, 69, 97, 99.
   a. (10 points) Draw a flow chart for the problem.
   b. (10 points) Write a C program for the problem.
   c. (40 points) Write an equivalent assembly program for the problem using ARMV7 Instructions Set Architecture (ISA). Also show the largest prime number in the seven-segment display. You must use directives such as if you have a data section or code section, show them clearly with directives.
   d. (20 points) Optimize your assembly code so that it takes as few instructions as possible.
   e. (20 points) Write equivalent machine instructions in hexadecimal format for each unique Instruction. (For example, if you have used an ADD instruction 7 times, you will write equivalent machine instruction for ADD once).